

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

May 2017

EC 402.01 RF & Microwave Engineering

Date: 23.05.2017, Tuesday Time: 1.30p.m. To 4.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

Q-1

- | | |
|--|---|
| (a) A space between two cavities in two cavity klystron is _____. | 1 |
| (b) In a quarter-wave transformer, input impedance is equal to _____. | 1 |
| (c) A reflex Klystron function as _____. | 1 |
| (d) Voltage reflection coefficient is the ratio of _____ wave to _____ wave. | 1 |
| (e) The Smith chart represents a family of circles obtained by varying normalized resistance in the range of _____. | 1 |
| (f) Klystron operates on the principle of _____. | 1 |
| (g) A two cavity klystron works as _____. | 1 |
| (h) Many circles are drawn in a smith chart used for transmission line calculations. The circles are shown in Fig below present _____. | 1 |



- | | |
|--|---|
| (i) Explain standing wave in brief. | 1 |
| (j) Why TEM mode cannot propagate in Waveguide? | 2 |
| Q-2 (a) Explain faraday rotation principle and faraday rotation isolator in detail | 6 |
| (b) Explain operation of magic tee. | 6 |

OR

- | | |
|---|----|
| Q-2 (a) Based on distribution theory of transmission line, draw equivalent circuit of transmission line and equation of voltage and current for lossless transmission line. Discuss standard assumptions. | 6 |
| (b) A rectangular air filled waveguide with dimension $7 \times 3.5 \text{ cm}^2$ cross is operated with a dominant mode $TE_{1,0}$ with frequency 3.5GHz. Find f_c, λ_g | 6 |
| Q -3 (a) Derive the wave equation with complete solutions regarding wave propagation in rectangular waveguide for TM mode. | 10 |
| (b) An air filled hollow rectangular waveguide has cross sections of $22.86 \times 10.16 \text{ cm}^2$. Calculate cutoff frequency for dominant mode? | 2 |

OR

- | | |
|--|----|
| Q-3 (a) Use Smith chart to find the following quantities for the transmission-line circuit which is connected with $Z_L = 100 + j100 \Omega$ and has $Z_0 = 50 \Omega$, length of line 0.2λ | 10 |
| (a) The SWR on the line. | |
| (b) The reflection coefficient at the load. | |
| (c) The input impedance of the line. | |
| (d) The distance from the load to the first voltage minimum. | |
| (e) The distance from the load to the first voltage maximum. | |
| (b) Design a quarter wave transmission transformers to match a line having impedance of 200 ohm, feeding at 30MHz to a load of 800ohm. | 2 |

Section –II

- Q-4(a)** Match the following **5**
- | X | Y |
|---|----------------------|
| (I) Hollow metal conducting pipes | (1) Cavity resonator |
| (II) interconnect two sections of waveguide | (2) circulator |
| (III) acts as a high-Q resonant circuit | (3) PN junction |
| (IV) three-port microwave device used for coupling energy in only one direction | (4) T section |
| (V) diodes is not typically used in the microwave region | (5) waveguide |
- (b)** Explain in detail electron bunching process with help of diagram. **3**
- (c)** What is pi mode jumping in magnetron? **2**
- (d)** What structure in the TWT delays the forward progress of the traveling wave? **1**
- Q- 5 (a)** Explain Gunn diode in detail. **6**
- (b)** What is tunneling? Explain energy band diagram and IV characteristics of tunnel diode. **6**
- OR**
- Q-5 (a)** Explain two cavity klystron amplifiers in detail. **6**
- (b)** Explain losses related to microstrip line. **2**
- (c)** A transmission line has characteristic impedance of 50 ohm and is terminated in a load impedance of 75 ohm. Calculate reflection coefficient and standing wave ratio. **4**
- Q- 6(a)** Explain microstrip line and strapline in brief. **6**
- (b)** Explain directional coupler in detail. **6**
- OR**
- Q-6 (a)** Explain circulator with the help of diagram. **6**
- (b)** Draw appropriate diagram of TWTA and explain amplification process along with diagram. **6**